Morphic Studio

Document 18 — AI Planner (Expanded Specification)

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Version: 2.0

Status: Expanded Specification

Priority: Highest (Core System)

Purpose

The AI Planner is the central intelligence and orchestration engine of Morphic Studio.

It is responsible for understanding creator intent, analyzing project context, planning production tasks, validating consistency, coordinating AI models, and ensuring every generation aligns with the project's Story Bible, Creative Memory Engine, Character Manager, World Builder, and Asset Library.

Unlike traditional AI systems that immediately generate output from a prompt, the AI Planner performs structured reasoning before generation.

The AI Planner never creates content blindly.

It plans first.

Vision

The AI Planner should behave like an experienced Creative Director.

Before production begins it asks:

- What is the creator trying to achieve?
- What already exists?
- What information is missing?
- What assets can be reused?

- What rules must be respected?
- Which AI models are best suited?
- What order should tasks happen?

Only after answering these questions should production begin.

Core Philosophy

Traditional AI:

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Prompt

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Generate
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Morphic Studio:

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Creator Request

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Understand

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Analyze

↓

Plan

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Validate

↓

Present Plan

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Creator Approval
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Execute

↓

Review

↓

Update Memory
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Planning is mandatory.

Objectives

The AI Planner shall:

- Understand creator intent.
- Analyze scripts.
- Detect missing information.
- Identify reusable assets.
- Build production plans.
- Coordinate AI systems.
- Prevent inconsistencies.
- Reduce unnecessary generation.
- Improve quality.
- Learn from creator decisions.

Responsibilities

The AI Planner is responsible for:

- Story analysis
- Character identification
- Timeline validation
- World validation
- Asset discovery
- AI model selection
- Task orchestration
- Workflow optimization

- Production scheduling
- Quality verification

Planning Workflow

Every request follows this workflow:

1. Receive creator request.
2. Read Story Bible.
3. Load Character Manager.
4. Load World Builder.
5. Search Asset Library.
6. Read Creative Memory Engine.
7. Detect missing information.
8. Build production plan.
9. Estimate required assets.
10. Recommend AI models.
11. Present plan.
12. Wait for creator approval.
13. Execute production.
14. Validate outputs.
15. Save approved assets.
16. Update project memory.

Context Collection

Before any generation the planner gathers:

Story Context

- Active chapter
- Active scene
- Story arc
- Narrative goals

Character Context

- Current appearance
- Active outfit
- Emotional state
- Relationships

- Recent development

World Context

- Current location
- Time of day
- Weather
- World rules
- Active organizations

Asset Context

- Existing references
- Previous generations
- Saved props
- Backgrounds
- Voice profiles
- Animations

Creative Context

The Creative Memory Engine provides:

- Emotional intent
- Artistic style
- Creator preferences
- Camera language
- Narrative pacing
- Visual symbolism
- Previous creative decisions

AI Model Selection

The planner decides which model performs each task.

Example:

Writing AI

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Image AI

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Animation AI

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Voice AI

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Audio AI

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Translation AI

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Quality Review AI

Models remain interchangeable.

The planner focuses on capability rather than vendor.

Task Graph

Rather than one large generation, production is divided into tasks.

Example:

Analyze Story

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Update Story Bible

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Generate Missing Character

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Generate Missing Outfit

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Generate Missing Background

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Create Storyboard

↓

Generate Comic Panels

↓

Review

↓

Creator Approval

↓

Finalize

Each task depends on previous results.

User Stories

As a creator, I want AI to understand my project before generating anything.

As a creator, I want unnecessary generations avoided.

As a creator, I want AI to explain why it recommends certain actions.

As a creator, I want complete control over every production plan.

Functional Requirements

The AI Planner shall:

- Analyze requests.
- Build execution plans.
- Detect missing assets.
- Recommend reusable assets.
- Select AI workflows.
- Estimate production time.
- Track progress.
- Pause workflows.
- Resume workflows.
- Cancel workflows.
- Learn creator preferences.

Explainability

Every recommendation should include an explanation.

Examples:

"I reused John's approved school uniform because it already exists in the Asset Library."

"A new location must be generated because none exists matching this description."

Transparency builds trust.

Database Entities

Core entities include:

- AIPlan
- ProductionTask
- TaskDependency
- PlannerDecision
- PlannerHistory
- AIWorkflow
- AIRecommendation
- ModelProfile

API Considerations

Potential endpoints:

- Create Plan
- Retrieve Plan
- Execute Plan
- Cancel Plan
- Resume Plan
- Planner Status
- Planner History
- AI Recommendation

Error Handling

If planning fails:

- Preserve project state.
- Explain the issue.
- Recommend corrective actions.
- Allow manual planning.

If execution fails:

- Retry individual tasks.
- Resume from checkpoints.
- Prevent duplicate work.

Non-Functional Requirements

The planner should:

- Respond quickly.
- Scale to complex productions.
- Support long-running workflows.
- Handle multiple AI providers.
- Cache reusable context.
- Minimize unnecessary API calls.

Future Expansion

Future versions may include:

- Automatic project scheduling.
- Budget estimation.
- GPU resource optimization.
- Distributed rendering.
- Multi-agent collaboration.
- Personalized planning strategies.
- Predictive production analysis.
- AI mentor mode.

Acceptance Criteria

The AI Planner succeeds when creators feel that Morphic Studio understands their project before creating anything, producing consistent, explainable, and efficient results.

Key Principle

The AI Planner is the brain of Morphic Studio. It does not create first—it understands, plans, and reasons before every creative action, ensuring that every output serves the creator's vision rather than guessing at it.

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